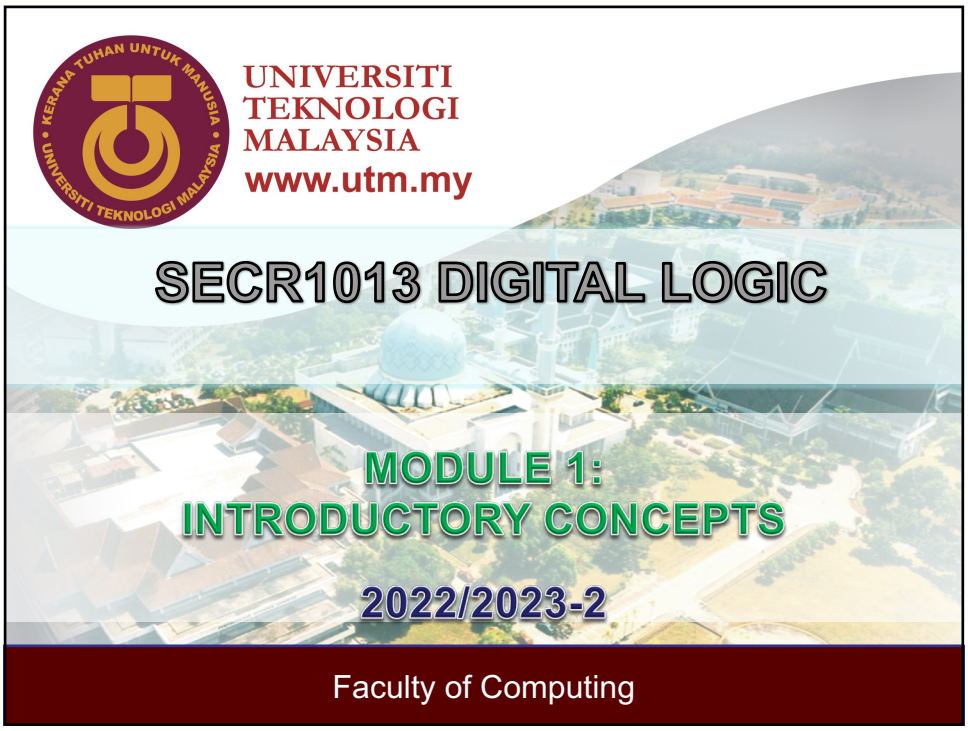
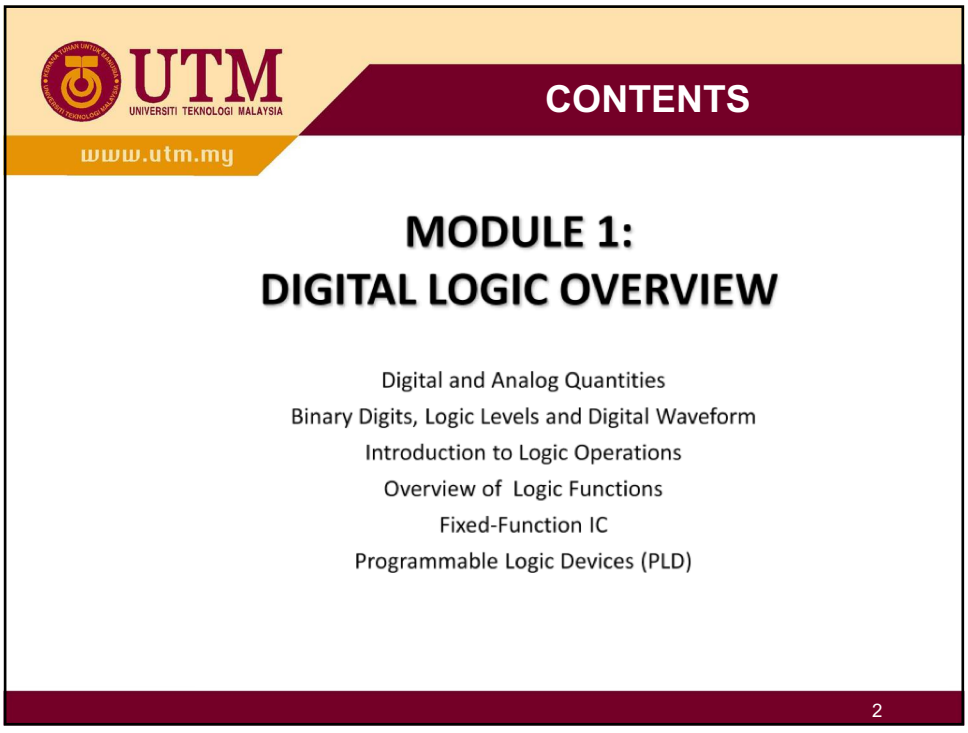
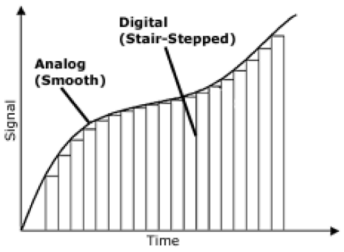

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MODULE 1:
INTRODUCTORY CONCEPTS
2022/2023-2
Faculty of Computing

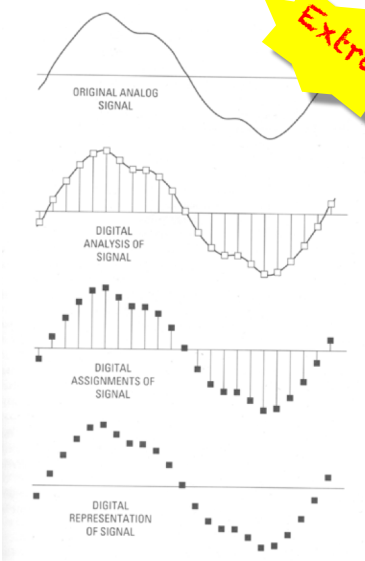

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CONTENTS
MODULE 1:
DIGITAL LOGIC OVERVIEW
Digital and Analog Quantities
Binary Digits, Logic Levels and Digital Waveform
Introduction to Logic Operations
Overview of Logic Functions
Fixed-Function IC
Programmable Logic Devices (PLD)
2

Example of sampling analog-to-digital (frequency at least 2 times higher than analog)



http://www.blazeaudio.com/howto/bg-digital.html

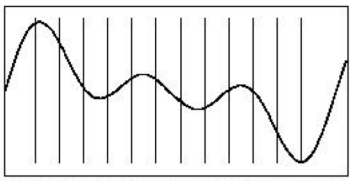
Extra



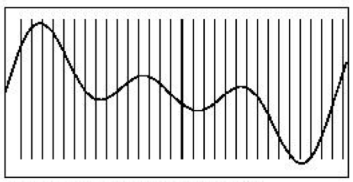
http://www.geardiary.com/2011/04/01/music-diary-notes-the-brave-new-world-of-digital-music/digital_sampling/

3

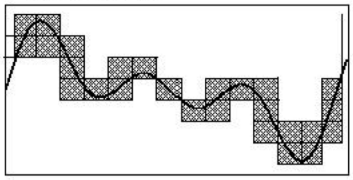
Extra



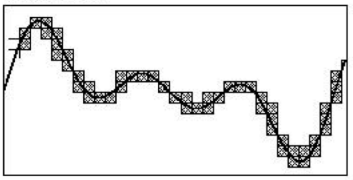
lower sample rates take fewer snapshots of the waveform .....



faster sample rates take more snapshots....



resulting in a rough recreation of the waveform.



resulting in a smoother and more detailed recreation of the waveform.

http://musicandcomputers306.blogspot.com/2010/10/waveforms-ad-conversion-sampling.html

4

Extra

Self-Test: Which of the following belong to analog system?


(a)


(b)


(c)


(d)


(e)


(f)

Resource: Google searched

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Digital

Exercise: Match the picture to which digital application system it belong to.





(a) Computers
(b) Manufacturing systems
(c) Medical Science
(d) Transportation
(e) Entertainment
(f) Telecommunications









Resource: http://www.wirelessnetworkproducts.com/dsl-2320b.aspx

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Exercise 1.1 : Calculate the frequency of signals if time period are given as the following:

a) $10ms =$ _____ Hz

b) $100\mu s =$ _____ KHz

c) $100ns =$ _____ MHz

d) $1000\,ps =$ _____ GHz

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Exercise 1.2 : Calculate the time period of signals if the frequencies are given as the following:

a) $1000KHz =$ _____ μs

b) $100MHz =$ _____ ns

c) $1000GHz =$ _____ ps

d) $100THz =$ _____ ps

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Exercise 1.3: Given the duration or period of a system is 1000ms, determine the *on state* and *off state* of the system that operate with the ratio of duty cycle is 25%. Show your works.

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Exercise 1.4: Given the duration or period of a system is 1000ms, determine the *on state* and *off state* of the system that operate with the ratio of duty cycle is 50%. Show your works.

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Exercise 1.5: Given the *duty cycles* of a system is 40% for a duration of a system is 500ms.

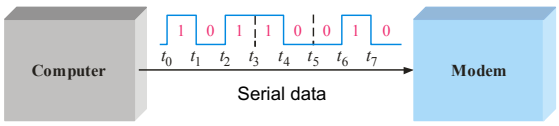
a) Calculate the pulse width of the system.
b) Determine the *off state* of the system that operate with the ratio of duty cycle.

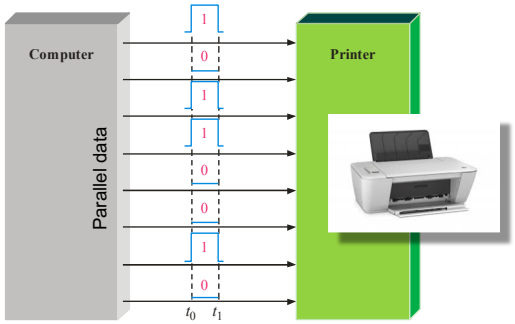
Show your works.

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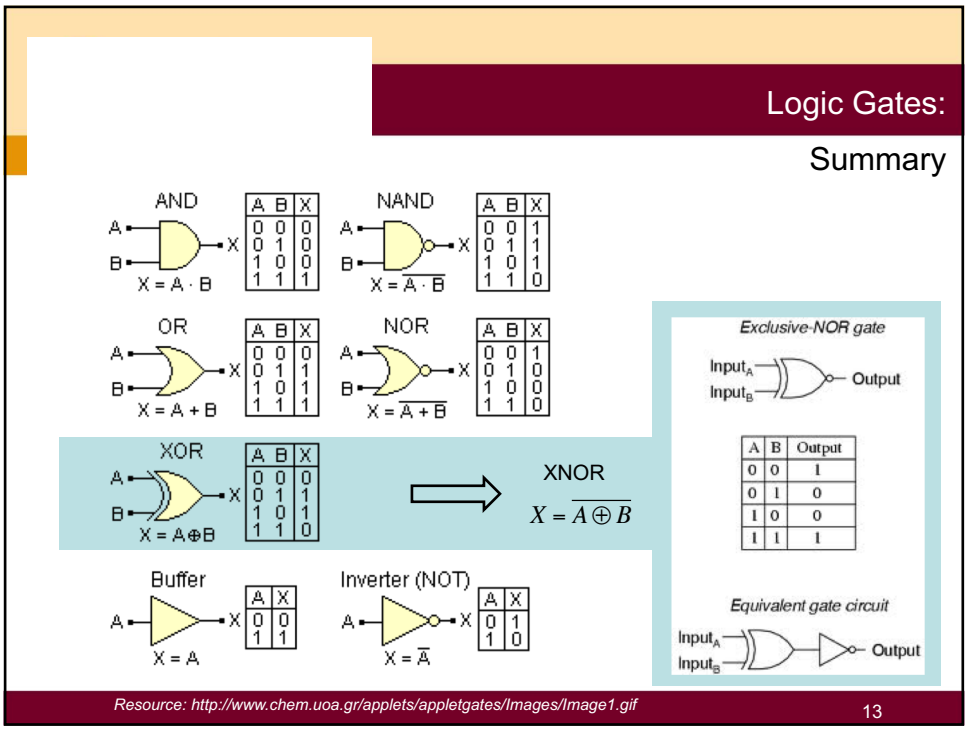
Data Transfer

Data can be transmitted by either **serial** transfer or **parallel** transfer.





Resource: Floyd, Digital Fundamentals, 10th Edition 12


Logic Gates:
Summary
AND
X = A · B
NAND
X = A · B
OR
X = A + B
NOR
X = A + B
XOR
X = A⊕B
XNOR
X = A ⊕ B
Buffer
X = A
Inverter (NOT)
X = A
Exclusive-NOR gate
Input A
Input B
Output
A B Output
0 0 1
0 1 0
1 0 0
1 1 1
Equivalent gate circuit
Input A
Input B
Output
Resource: http://www.chem.uoa.gr/applets/appletgates/Images/Image1.gif
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